

# Exploratory Data Analysis Report

Project 2 — Business Intelligence Report

|                                                |                                                       |
|------------------------------------------------|-------------------------------------------------------|
| Prepared by<br><b>Nada Zohny</b>               | Organization<br>-                                     |
| Dataset<br><b>Sales Orders — 1,200 Records</b> | Report Type<br><b>Exploratory Data Analysis (EDA)</b> |

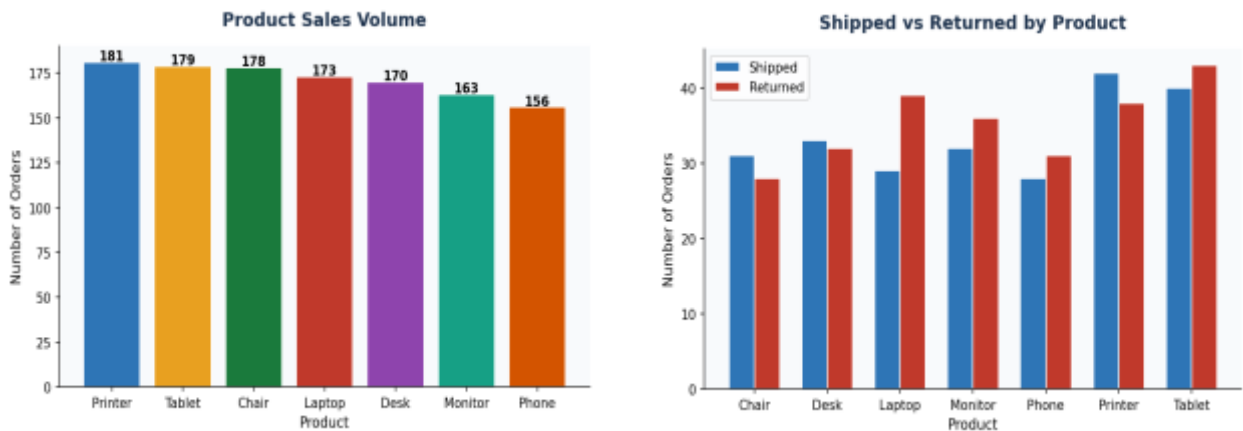
*This report presents a comprehensive exploratory analysis of a sales order dataset comprising 1,200 transactions across 14 variables. The analysis uncovers patterns in product performance, customer retention, referral source effectiveness, and seasonal trends — translating raw data into actionable business intelligence.*



## Section 1: Product Performance

### 1.1 Overview

This section examines how different products perform across key metrics including total sales volume, shipping success, and return rates. Understanding product performance helps the business identify its strongest revenue drivers and most problematic items.



### 1.2 Key Findings

| Metric                | Finding                        | Value   |
|-----------------------|--------------------------------|---------|
| Most Sold Product     | Printer leads all categories   | Printer |
| Most Shipped Product  | Printer dominates fulfilment   | Printer |
| Most Returned Product | Tablet has highest return rate | Tablet  |

### 1.3 Problem & Bright Side

#### Problem

The Tablet product has the highest return rate, which signals potential quality issues, misleading product descriptions, or unmet customer expectations. High return rates increase operational costs significantly.

#### Bright Side

Printer is both the most sold and most shipped product, indicating strong demand and reliable fulfilment. This product line is a clear revenue anchor for the business.

# 1.4 Responsible Department & Solution Plan

Responsible Department: Product Management & Quality Assurance

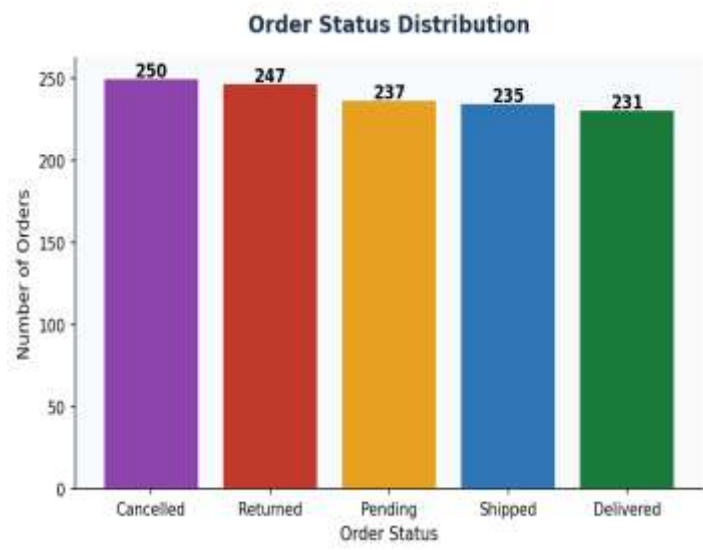
| Month     | Action                                                      | Owner          | Goal                           |
|-----------|-------------------------------------------------------------|----------------|--------------------------------|
| Month 1-2 | Conduct Tablet quality audit and gather return reason data  | Product Team   | Identify root cause of returns |
| Month 3-4 | Update product descriptions and add detailed specifications | Marketing Team | Reduce expectation mismatch    |
| Month 5-6 | Launch Printer bundle promotions to capitalize on demand    | Sales Team     | Increase average order value   |



## Section 2: Order Status Breakdown

### 2.1 Overview

This section analyses the distribution of orders across five statuses: Shipped, Delivered, Cancelled, Returned, and Pending. A healthy business should have the majority of orders reaching Delivered status with minimal Cancellations and Returns.



### 2.2 Key Findings

| Status    | Count | % of Total |
|-----------|-------|------------|
| Shipped   | 235   | 19.6%      |
| Delivered | 231   | 19.3%      |
| Cancelled | 250   | 20.8%      |
| Returned  | 247   | 20.6%      |
| Pending   | 237   | 19.8%      |

### 2.3 Problem & Bright Side

**Problem**

Cancelled (250) and Returned (247) orders combined account for 497 transactions — 41.4% of all orders. This is an alarming rate that directly impacts revenue, operational costs, and customer satisfaction. Only 19.3% of orders are fully delivered.

### Bright Side

Orders are almost evenly distributed across all statuses, which suggests the pipeline is active at all stages. The Shipped count (235) is healthy, indicating the fulfilment process itself is functioning.

## 2.4 Responsible Department & Solution Plan

Responsible Department: Operations, Customer Service & Logistics

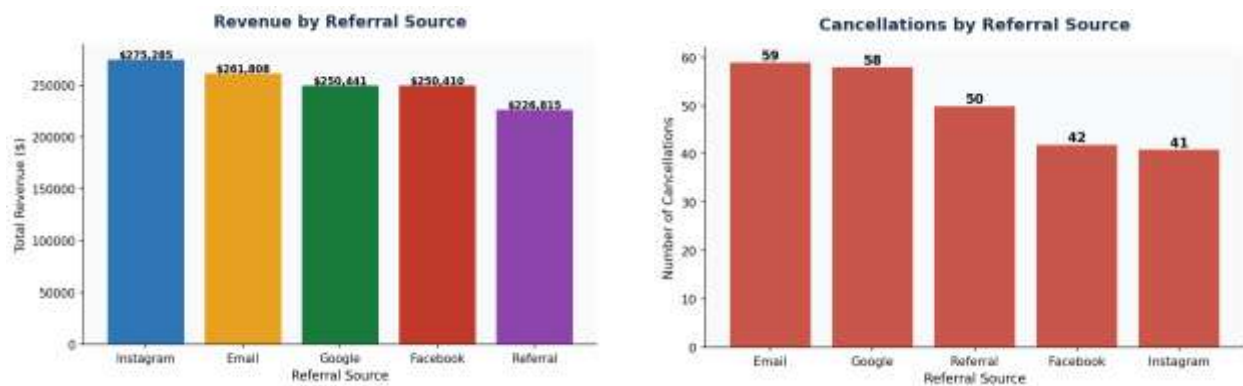
| Month     | Action                                                                  | Owner            | Goal                            |
|-----------|-------------------------------------------------------------------------|------------------|---------------------------------|
| Month 1-2 | Investigate top cancellation reasons via customer surveys               | Customer Service | Identify cancellation triggers  |
| Month 3-4 | Implement order confirmation reminders and easier modification options  | Operations       | Reduce cancellation rate by 20% |
| Month 5-6 | Launch post-delivery follow-up to reduce returns through better support | Customer Service | Reduce return rate by 15%       |



## Section 3: Referral Source Analysis

### 3.1 Overview

This section evaluates how each marketing and referral channel performs in terms of customer acquisition, revenue generation, and negative outcomes such as cancellations. Understanding channel performance helps the business allocate marketing budgets more effectively.



### 3.2 Key Findings

| Metric                 | Source    | Implication                  |
|------------------------|-----------|------------------------------|
| Most Popular Source    | Instagram | Highest customer acquisition |
| Highest Revenue Source | Instagram | Best ROI channel             |
| Most Cancellations     | Email     | Lowest quality leads         |
| Most Coupon Users      | Instagram | High engagement channel      |

### 3.3 Problem & Bright Side

#### Problem

Email campaigns generate the most cancellations, suggesting that email-acquired customers have lower purchase intent or are attracted by misleading promotions. This wastes marketing resources and inflates cancellation rates.

#### Bright Side

Instagram is the top performer across all positive metrics — most customers, highest revenue, and most coupon engagement. This channel has proven to attract high-intent buyers and should receive increased investment.

### 3.4 Responsible Department & Solution Plan

Responsible Department: Digital Marketing & Growth Team

| Month     | Action                                                            | Owner           | Goal                              |
|-----------|-------------------------------------------------------------------|-----------------|-----------------------------------|
| Month 1-2 | Increase Instagram ad budget by 25% and test new creative formats | Marketing Team  | Grow Instagram-acquired customers |
| Month 3-4 | Audit and redesign email campaigns for clarity and honesty        | Email Marketing | Reduce email cancellations by 30% |
| Month 5-6 | Launch Instagram-exclusive loyalty program with coupon rewards    | Growth Team     | Improve Instagram retention rate  |



## Section 4: Coupon Analysis

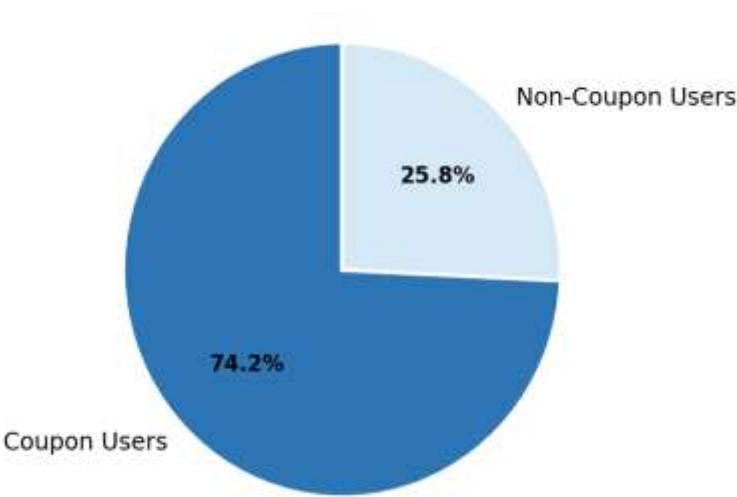
### 4.1 Overview

This section examines coupon usage patterns across the customer base. Coupons are a key marketing tool and understanding their adoption rate, combined with referral source data, helps evaluate promotional strategy effectiveness.

### 4.2 Key Findings

| Metric                     | Value     | Percentage              |
|----------------------------|-----------|-------------------------|
| Coupon Users               | 891       | 74.3%                   |
| Non-Coupon Users           | 309       | 25.8%                   |
| Most Active Coupon Channel | Instagram | Highest coupon adoption |

Coupon Users vs Non-Coupon Users



### 4.3 Problem & Bright Side

Problem



74.3% of customers require a coupon to complete a purchase. This indicates heavy price sensitivity and potential over-reliance on discounts. If coupons are removed, a large portion of customers may not purchase at full price, reducing profit margins.

### Bright Side

High coupon adoption shows strong promotional campaign reach. Coupons are clearly driving conversions and can be strategically used to target specific segments or time periods like slow months (September).

## 4.4 Responsible Department & Solution Plan

Responsible Department: Pricing Strategy & Marketing

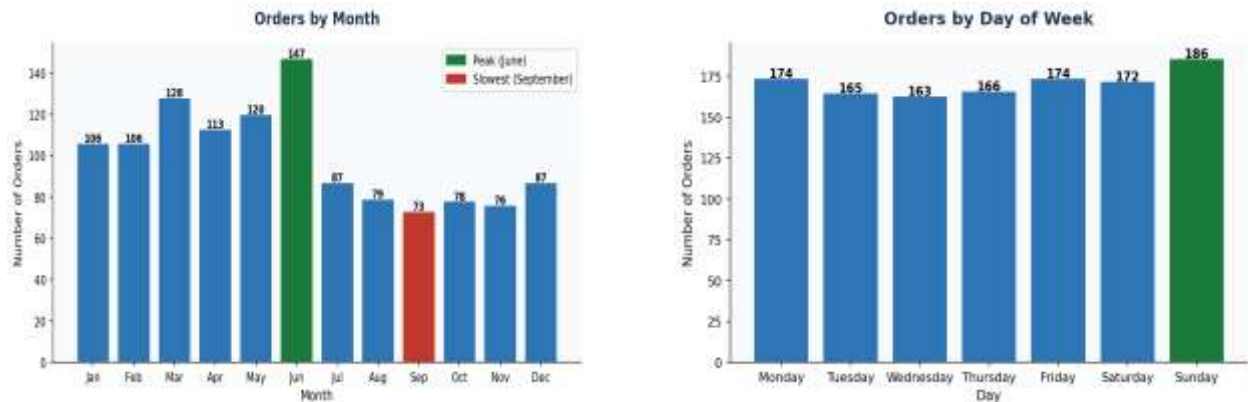
| Month     | Action                                                                | Owner             | Goal                              |
|-----------|-----------------------------------------------------------------------|-------------------|-----------------------------------|
| Month 1-2 | Analyze profit margins of coupon vs non-coupon orders                 | Finance Team      | Understand true coupon cost       |
| Month 3-4 | Introduce tiered loyalty points as alternative to blanket discounts   | Marketing Team    | Reduce discount dependency        |
| Month 5-6 | Test full-price promotions with value-add offers instead of discounts | Product Marketing | Increase full-price purchase rate |



## Section 5: Time Analysis

### 5.1 Overview

This section identifies when customers are most and least active. Understanding temporal patterns allows the business to align marketing campaigns, staffing, and promotions with peak and off-peak periods.



### 5.2 Key Findings

| Metric        | Finding   | Implication                |
|---------------|-----------|----------------------------|
| Busiest Day   | Sunday    | Customers shop on weekends |
| Busiest Month | June      | Summer peak season         |
| Slowest Month | September | Post-summer decline        |

### 5.3 Problem & Bright Side

#### Problem

September shows a significant drop in orders, representing a predictable but unaddressed revenue gap. Without targeted intervention, the business loses potential revenue every year during this period.

#### Bright Side

Sunday is the busiest day and June is the peak month — both are predictable patterns the business can plan around. Knowing peak times allows for proactive inventory preparation and targeted promotions.

## 5.4 Responsible Department & Solution Plan

Responsible Department: Marketing & Inventory Management

| Month     | Action                                                              | Owner          | Goal                          |
|-----------|---------------------------------------------------------------------|----------------|-------------------------------|
| Month 1-2 | Plan June campaign calendar in advance with special offers          | Marketing Team | Maximize peak season revenue  |
| Month 3-4 | Design September re-engagement campaign targeting dormant customers | CRM Team       | Reduce September revenue drop |
| Month 5-6 | Launch Sunday-exclusive flash deals and weekend promotions          | Marketing Team | Increase weekend basket size  |

## Section 6: Customer Retention

### 6.1 Overview

This section examines customer loyalty and repeat purchase behaviour. Customer retention is one of the most critical metrics for long-term business health, as retaining an existing customer is significantly cheaper than acquiring a new one.

**Customer Retention: One-Time vs Returning**



### 6.2 Key Findings

| Metric                            | Value  | Implication                   |
|-----------------------------------|--------|-------------------------------|
| Total Customers                   | 1,200  | Full customer base            |
| Returning Customers               | 11     | Only 0.9% retention rate      |
| Max Orders by One Customer        | 6      | Highest loyalty observed      |
| Top Product — Returning Customers | Tablet | Loyal customers prefer Tablet |

### 6.3 Problem & Bright Side

**Problem**

A customer retention rate of only 0.9% (11 out of 1,200 customers) is critically low. This means the business is almost entirely dependent on new customer acquisition to survive. If new customer flow slows down, revenue will collapse immediately.

### Bright Side

The 11 returning customers demonstrate that repeat purchasing is possible. The customer who placed 6 orders proves strong loyalty can be built. These customers can serve as a model for what an ideal customer journey looks like.

## 6.4 Responsible Department & Solution Plan

Responsible Department: CRM, Customer Success & Marketing

| Month     | Action                                                                   | Owner             | Goal                           |
|-----------|--------------------------------------------------------------------------|-------------------|--------------------------------|
| Month 1-2 | Interview the 11 returning customers to understand their motivation      | CRM Team          | Build retention persona        |
| Month 3-4 | Launch a post-purchase email sequence with personalized follow-up offers | Email Marketing   | Convert 5% of one-time buyers  |
| Month 5-6 | Introduce a loyalty rewards program with points per purchase             | Product Marketing | Increase retention rate to 5%+ |



## Conclusion & Strategic Recommendations

### Executive Summary

This exploratory data analysis of 1,200 sales orders has revealed several critical business insights. While the business shows strong product demand and effective use of Instagram as a marketing channel, it faces serious challenges in customer retention, order fulfilment, and over-reliance on discount pricing.

### Top 5 Key Findings

- Customer retention is critically low at 0.9% — the most urgent issue requiring immediate action.
- Cancellations and Returns combined account for 41.4% of all orders — a major revenue and operational drain.
- Instagram is the top-performing channel across all positive metrics and should receive increased investment.
- 74.3% of customers require coupons — indicating heavy price sensitivity and discount dependency.
- June is peak season and September is the slowest month — predictable patterns that must be planned for.

### Integrated 6-Month Solution Plan

The following plan integrates all departmental recommendations into a unified timeline:

| Phase     | Focus Area   | Key Action                                                                | Owner           | Success Metric            |
|-----------|--------------|---------------------------------------------------------------------------|-----------------|---------------------------|
| Month 1-2 | Diagnosis    | Audit returns, survey cancellations, interview loyal customers            | Product + CRM   | Root causes identified    |
| Month 3-4 | Optimization | Fix product listings, redesign email campaigns, increase Instagram budget | Marketing + Ops | 20% drop in cancellations |
| Month 5-6 | Growth       | Launch loyalty program, September campaign, Sunday flash deals            | CRM + Marketing | Retention rate above 5%   |

## Final Note

All insights are derived from the provided sales dataset of 1,200 records. The recommendations are data-driven and designed to be actionable within a 6-month implementation window.